

lecture 01

foundations

course overview // logarithms; derivatives; integrals; matrices; OLS

weekly assessment

a practice run

course overview

As an aspiring academic, you must establish a
reputation of excellence.

What are our goals?

What are our goals?

Use, understand, and explain
the **tools** of modern statistical modeling.

~~paint mixing~~

engines	tools to fit models	ML, MM, Bayes, EM
distributions	tools to model the world	normal, bernoulli, Poisson
quantities of interest	tools to understand models	expected value, first difference
measures of fit	tools to evaluate fitted models	predictive distributions, LOOIC
tests and CIs	tools to evaluate hypotheses	Wald, LR, score
properties	tools to evaluate procedures	bias, consistency, coverage
computing	tools to compute estimates	glm(); {brms}

What do I ask of you?

What do I ask of you?

1. Read the notes and complete the exercises.
2. Write a good paper of “modest ambition” and “exceptional quality.”
3. Run a good workshop.
4. Talk to me about all this stuff.

do you understand a variety of tools?

weekly assessments,	10%	in-class, closed book assessments based on the examples and exercises.
midterm exam	15%	
<i>and</i> final exam	25%	

can you explain these tools simply, clearly, and correctly?

workshop	15%	one-hour workshop on a particular statistical model
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can you apply these tools well?

paper	35%	research project of modest ambition and exceptional quality
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Exams

1. Based closely on exercises and examples.
 - Variants are fair game.
 - Make sure you can comfortably complete examples and exercises in a closed-book setting.
 - Make sure you are not relying on solutions or AI-help. Don't confuse "familiarity" with "understanding."
2. Exercises are "due" weekly, but I do not grade them. (The weekly assessment comes from the exercises!)
 - I'll distribute detailed solutions.
 - Come talk to me (or our TA) often!
 - Collaborate as it helps you learn. Ideal model might be (1) make a best effort as an individual, (2) discuss difficult portions in groups of 2-3, (3) polish your work as an individual, and (4) review your work weekly.
 - Large group discussion? TA-led study group?

Research Project

Your task this semester: write a paper of **modest ambition** and **exceptional quality** that makes a **descriptive argument** about politics.

“Descriptive Argument”

“A descriptive argument describes some aspect of the world. In doing so, it aims to answer ‘what’ questions (e.g., when, whom, out of what, in what manner) about a phenomenon or a set of phenomena. Descriptive arguments are about what is/was. For example: ‘Over the course of the past two centuries there have been three major waves of democratization.’” —Gerring (2012)

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Mere Description

JOHN GERRING*

This article attempts to reformulate and resuscitate the seemingly prosaic methodological task of description, which is often derided in favour of causal analysis. First, the problem of definition is addressed: what does this category of analysis (‘description’) refer to? Secondly, a taxonomy of descriptive arguments is offered, emphasizing the diversity contained within this genre of empirical analysis. Thirdly, the demise of description within political science is charted over the past century, with comparisons to other disciplines. Fourthly, it is argued that the task of description ought to be approached independently, not merely as a handmaiden of causal theories. Fifthly, the methodological difficulties of descriptive inference are addressed. Finally, fruitful research areas within the rubric of description are reviewed.

‘What the devil is going on around here?’

Abraham Kaplan¹

In recent times, the quest for scientific understanding in the social sciences has come to be equated with a causal understanding of the world. The task of description, by contrast, is often identified with idiographic storytelling or with messy, observational data that is insufficient to reach causal inference. Indeed, the term has come to be employed as a euphemism for a failed, or not yet proven, causal inference. Studies that do not engage causal or predictive questions, or do not do so successfully, are judged ‘merely’ descriptive. The implication is that description is a mundane task – necessary, to be sure, and occasionally quite complex, but of little intrinsic scientific value.²

To be sure, research continues on a variety of well-established topics and advances in several areas are impressive.³ Yet, despite these achievements, description has never been recognized as a methodological topic *per se*, as has causation. There are, for example, virtually no journals, articles, books or courses devoted to the subject. Instead, one finds a

* Department of Political Science, Boston University (email: jgerring@bu.edu). This article has benefited enormously from comments received from Robert Adcock, Ben Bishin, Fred Chernoff, Michael Coppedge, Zachary Elkins, Colin Elman, Gary Goertz, Andy Harris, Patrick Johnston, Evan Lieberman, Drew Linzer, James Mahoney, Fred Schaffer, Andreas Schedler, Carsten Schneider and David Waldner. The author is also grateful to Joshua Yenowicz, who conducted the content analysis for Figures 2–4.

¹ Abraham Kaplan, *The Conduct of Inquiry: Methodology for Behavioral Science* (San Francisco: Chandler, 1964), p. 85.

² It is not clear when, precisely, this pejorative connotation arose. It was invoked, or commented on, in the social science literature at various points in the mid-to-late twentieth century (e.g., Lester E. Klimm, ‘Mere Description’, *Economic Geography*, 35:1 (1959), unnumbered; Amartya Sen, ‘Description as Choice’, *Oxford Economic Papers*, 32 (1980), 353–69; J. David Singer, ‘The Level-of-Analysis Problem in International Relations’, *World Politics*, 14 (1961), 77–92). However, it probably stretches back further in time within the tradition of Anglo-American economics and political science (e.g., John Clark and Joseph Banks, ‘Description of an Extraordinary Production of Human Generation, with Observations’, *Philosophical Transactions of the Royal Society of London*, 83 (1793), 154–63, p. 157).

³ These areas are reviewed briefly in the concluding section of this article.

The Profession

A Case for Description

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ABSTRACT Descriptive research—work aimed at answering “who,” “what,” “when,” “where,” and “how” questions—is vital at every stage of social scientific inquiry. The creative and analytic process of description—through concepts, measures, or cases, whether in numeric or narrative form—is crucial for conducting research aimed at understanding politics in action. Yet, our field tends to devalue such work as “merely descriptive” (Gerring 2012), subsidiary to or less valuable than hypothesis-drive causal inference. This article posits four key areas in which description contributes to political science: in conceptualization, in policy relevance, in the management and leveraging of data, and in challenging entrenched biases and diversifying our field.

The emphasis on a “more scientific” understanding of politics, associated with generalization and parsimony, that has dominated the discipline since the behavioral revolution has privileged a causal approach to understanding the social world. Conversely, descriptive inquiry has a diminished role and status in the discipline. It often is regarded as messier, less important, or ancillary. Yet, scholars across methodological, epistemological, and subfield divides benefit from engaging in descriptive work. In response, we organized a series of collaborative workshops and roundtables focused on description as a social science enterprise. These gatherings served as a basis for discussing what “counts” as description in social science and what it contributes to the research process. By sharing insights from our collaborations, this article examines the contributions of descriptive research and presents an argument for its importance, in regard to both hypothesis-driven inquiry and as a standalone strategy and method.

Separate from the emphasis of causal research, descriptive research aims to answer “who,” “what,” “when,” “where,” and “how” questions.¹ It is both a creative and an analytic process. Although “describing often involves practices which, like copying

and translating, [which] tend...to be associated with lack of originality and novelty, description is neither neutral nor passive, nor does it simply entail transcribing or rearticulating the given” (Weinstein 2022). This is because description involves making choices in the categorizing, sense making, and management of data.

Given the central role of description in political science, a brief discussion of its history (and eventual marginalization) is important. The behavioral revolution in political science was articulated by Merriam (1926) as an approach closely aligned with the natural sciences by Key as an effort to make research “more effective” (1950, 20); by Truman (1951, 37) as an “emphasis on empirical methods...[in] the development of a science of the political process”; and by Dahl (1951, 768) as “the modern scientific outlook.” It was a Kuhnian paradigm shift in the study of politics. Evolving through and with the expansion of computing technology in the mid- to late-twentieth century, the behavioral turn in political science is deeply linked with the application of empirical methods to data to yield causal arguments. This approach was consistent through the elaboration of the natural science model in works such as *Designing Social Inquiry* (King, Keohane, and Verba 1994) as well as in the Perestroika movement, which urged methodological pluralism (Monroe 2002) but did not question the underlying need for predominantly hypothesis-driven causal arguments.

This emphasis on causality has led to a decline in the prevalence of descriptive research in our field, which can be observed through the decrease of descriptive articles published in major field journals or in the phrasing that often accompanies such work

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Good Description*

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Anna Grzymala-Busse[‡]

May 19, 2025

Abstract

What distinguishes ‘good’ description from ‘mere’ description? We propose a framework for evaluating descriptive empirical social science, premised on the idea that descriptive research – like all empirical social science – should be grounded in theory. First, we articulate the social scientific purpose of description, which provides criteria for assessing descriptive questions. Good descriptive questions seek to uncover facts in need of explanation, build theory, or help to evaluate and revise theory. Second, we articulate three ideal characteristics of descriptive analyses. Good descriptive analyses are clear, comparable, and complete. Such analyses should be tightly and transparently linked to specific research questions (and thus theory), well contextualized, and as comprehensive – in their measurement and specification – as possible.

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“Modest Ambition”

As a starting point, imagine 3,000 words that make a single, specific, narrow contribution.

Hint: imagine something that will take about 3 weeks to put together.

Carlisle's AHH Scale

1. A body of work worthy of tenure at a major research university.
2. A book-length project.
3. An ambitious full-length article aimed at a top journal (e.g., AJPS or IO).
4. A full-length article aimed at a good subfield journal (e.g., POBE or JCR).
5. A research note aimed at a good journal (e.g., RAP).

“Exceptional Quality”

Hi Prof. Rainey,

I’ve attached a draft of my paper on [topic]. I’m thinking about submitting it to [journal] soon. Could you let me know if you have any suggested revisions prior to submission?

Grade	As a dissertation advisor, I’d say...
100%	“This seems ready to submit to a peer-reviewed journal. You should present it at a department workshop to help tie up any loose ends, but it seems ready to go.”
90%	“This seems <i>almost</i> ready to submit to a peer-reviewed journal. You should fix the minor issues listed below and then present it at a department workshop to help tie up any loose ends. But once you’ve fixed these minor issues, it seems ready to go.”
80%	“This is a good start. I see a couple of significant issues (and a few minor things) listed below that need work. Once you’ve addressed these issues, send me an updated version and I’ll give it another look.”
70%	“This paper isn’t adequately polished. Papers submitted to journals are significantly more polished than this—you should spend a lot of time revising and clarifying your paper before submitting. I’ve included some main areas to focus on below. In the future, it’s fine to discuss early work with me or ask me to review sketches of ideas, but it isn’t a good use of my time (or yours) to ask me for comments on a paper that you haven’t spend significant time making as good as you can. For drafts this rough, the arguments are tough to understand (and therefore critique and improve).

Papers that fail to demonstrate competence with the methods are graded less than 70%.

Workshop

Your task this semester: **design and run a 60-minute workshop on a particular statistical model.** (Ideally in teams of two.)

Required materials

1. Take-home handout.
2. Slides.
3. A read-ahead document.

Workshop structure

1. 15-minute lecture.
2. 15-minute example.
3. 20-minute guided practice.

Aggressive Scaffolding

Date	Title	Exam	Research Project	Workshop Due
Aug 25	Review	—	—	—
Sep 01	ML	—	Optional	—
Sep 08	SE	—	Optional	—
Sep 15	Covariates	—	Optional	—
Sep 22	{marginaleffects}	—	Optional	—
Sep 29	Bayes	—	Prospectus	—
Oct 06	MCMC	—	—	—
Oct 13	—	Midterm	—	—
Oct 20	AI	—	First draft	Initial outline of handout
Oct 27	Hierarchical Models	—	Peer Review Memo	First drafts of all materials
Nov 03	Hierarchical Models II	—	—	Workshop rehearsal
Nov 10	IRT	—	—	Hold workshop
Nov 17	A Final Perspective	—	—	
Nov 24	Wrap	—	Second draft	
Dec 01	Posters	—	Poster presentation	—
Dec 08	—	Final	Final draft	—

foundations

logarithms

Simplify $\log \left(\prod_{i=1}^n y_i^{a_i} \right)$.

derivatives

Let $\ell(\pi) = S \log(\pi) + (n - S) \log(1 - \pi)$ for $0 < \pi < 1$, where S and n (with $0 \leq S \leq n$) are fixed numerical constants. Find $\frac{d\ell(\pi)}{d\pi}$.

matrices

Compute the product of the 2×2 matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and
the 2×3 matrix $B = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}$.

integrals, pmfs , and pdfs

Compute $\Pr(-1.96 \leq X \leq 1.96)$ for $X \sim \mathcal{N}(0, 1)$.

expected values and
variances

Suppose the regression model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \epsilon_i.$$

1. Find the marginal effect of x .
2. Show that if the coefficient estimates are unbiased, then the plug-in marginal effect estimate is unbiased. (Would this generalize to, say $\frac{\beta_1}{\beta_3}$?)
3. Find the variance of the plug-in marginal effect estimate.

OLS and R

Starting with the code to the right, reproduce Clark and Golder's (2006) 1946-2000 Established Democracies model in Table 2 on p. 698. See also eq. 4 on p. 695 for their model specification. They use HC1 cluster-robust standard errors.

```
# run just once to get data package, then delete
devtools::install_github("carlislerainey/crdata")

# load packages
library(tidyverse)
library(sandwich) # for robust SEs

# load Clark and Golder's data
cg <- crdata::cg2006 # from my data package

# quick look at variables and their names
glimpse(cg)
```

Table 2
The Strategic Modifying Effect of Electoral Laws

Regressor	Dependent Variable: Effective Number of Electoral Parties									
	Cross-Sectional Analysis						Pooled Analysis			
	1980s Amorim Neto & Cox Data ^a	1980s Amorim Neto & Cox Data ^a	1990s Whole Sample	1990s Established Democracies ^b	1946 to 2000 Whole Sample	1946 to 2000 Established Democracies ^b				
Ethnic		-0.05 (0.28)	0.06 (0.37)	-0.70 (0.68)	0.19 (0.13)	0.11 (0.14)				
ln(Magnitude)		-0.08 (0.30)	0.51 (0.44)	-0.61 (0.59)	0.33* (0.20)	0.08 (0.23)				
UppertierSeats	0.04** (0.01)	-0.07 (0.04)	0.01 (0.02)	-0.02 (0.06)	0.05*** (0.02)	-0.06* (0.03)				
PresidentCandidates		0.22 (0.27)	0.36 (0.26)	0.07 (0.22)	0.35** (0.16)	0.26* (0.15)				
Proximity	-6.05*** (0.88)	-5.88*** (0.84)	-4.19*** (1.26)	-4.95*** (1.24)	-3.42*** (0.55)	-3.10*** (0.46)				
Ethnic × ln(Magnitude)	0.39*** (0.07)	0.37* (0.20)	-0.09 (0.17)	0.63* (0.34)	0.08 (0.12)	0.26 (0.17)				
Ethnic × UppertierSeats		0.07*** (0.02)	-0.005 (0.01)	0.01 (0.04)	-0.02** (0.01)	0.06*** (0.02)				
PresidentCandidates × Proximity	2.09*** (0.26)	1.84*** (0.43)	0.99** (0.46)	1.42*** (0.44)	0.80*** (0.23)	0.68*** (0.23)				
Constant	2.40*** (0.21)	2.60*** (0.51)	4.08*** (0.95)	5.15*** (1.32)	2.81*** (0.34)	2.92*** (0.35)				
Observations	51	51	62	39	555	487				
R ²	.71	.77	.29	.48	.30	.40				

Note: Standard errors are given in parentheses for cross-sectional models; robust standard errors clustered by country are used for the pooled models.

a. See Amorim Neto and Cox (1997).

b. Established Democracies omits elections from countries that transitioned to democracy after 1989.

* $p < .10$. ** $p < .05$. *** $p < .01$.

time permitting...

Show that scalar notation

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \epsilon_i$$

is equivalent to the matrix notation

$$y = X\beta + \epsilon.$$

what we'll learn

extensions to linear regression model

several methods to fit the models

generic methods to compute quantities of interest

my view is modest

these tools are **not** magic

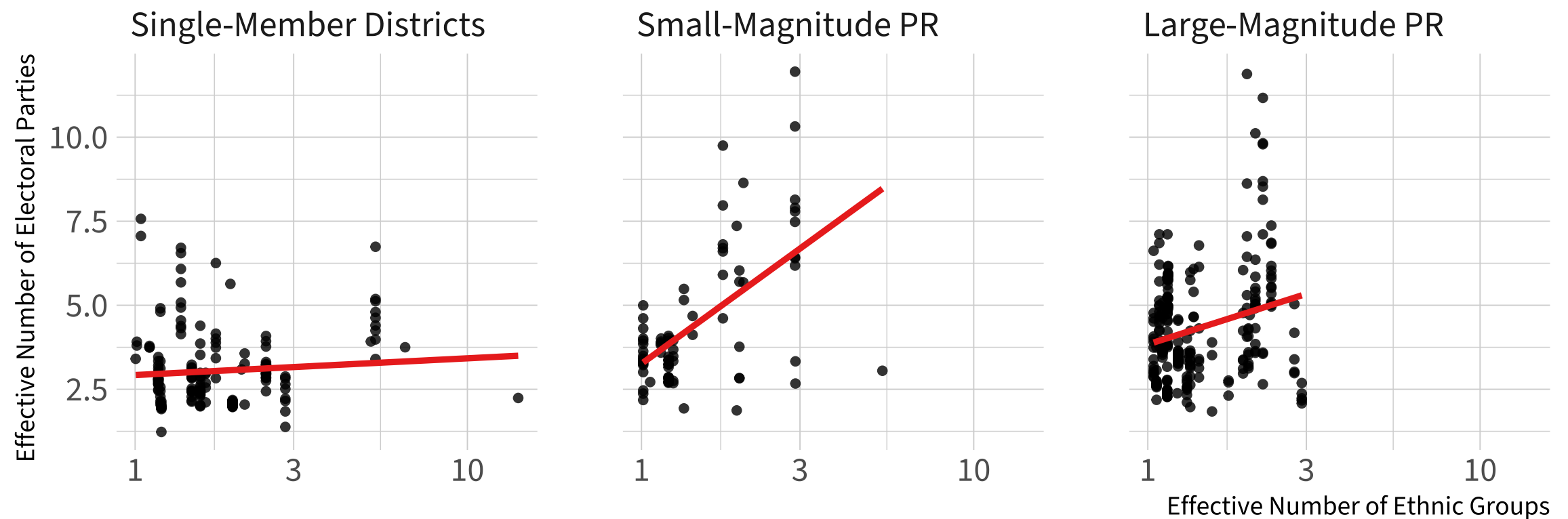
these tools are *modest extensions*
of scatterplots and histograms

We're almost always interested in $E(y \mid x)$
or $E(y \mid x = \text{hi}) - E(y \mid x = \text{lo})$.

*Hypothesis 1: Social heterogeneity increases the number of parties, but only when electoral institutions are sufficiently permissive.*⁴

A Simple Scatterplot

It's hard to do a lot better than this.



next time

$$y_i \sim f(y_i \mid \mu, \phi)$$

$$g(\mu) = \theta$$

Week 1 Exercises

Week 1 Exercises

I created some notes that review the concepts required for these exercises, but you should draw on familiar textbooks and notes where possible.

Do problems 4(f), 4(g), 5, 7, 8, 9, 10, 13(1), 14, 15, 20, 21, 22, 24(c), 24(g), 25(b), 25(e), 26(c), and 27. Use the other exercises as needed to brush up on weaknesses.

1 Fractions, Logarithms, and Exponents

Exercise 1 Some Practice with Fractions

Simplify each of the following:

a. $\frac{x}{y} + \frac{z}{y}$

b. $\frac{m}{n} - \frac{r}{n}$

c. $\frac{p+q}{r} + \frac{s}{r}$

d. $\frac{2a}{b} + \frac{3a}{b}$

e. $\frac{x}{2} + \frac{y}{3}$

f. $\frac{1}{a} - \frac{1}{b}$ (write as a single fraction)

Optional Early Assignments

- **Descriptive Research in Your Field** (*Optional; due by Sep. 1*) Reflect on what descriptive research looks like in your field and identify examples of high-quality description. Consider the following prompts as starting points:
 - a. Engage with Gerring (2012), de Kadt and Grzymala-Busse (2025), and Holmes et al. (2024).² What parts of their arguments do you find most compelling?
 - b. What is the value of description in your research area? Take one or two papers that you admire—the kinds of papers that you’d like to write—and ask yourself “what would this project have looked like if the authors had instead focused on carefully *describing* their critical concepts (rather than looking for their effects and/or causes)?”
- **Data Sets** (*Optional; due by Sep. 8*) Identify publicly available data sets relevant to your general area of interest. For each data set, briefly describe the variables, cases, and time periods included. Flag any variables that seem especially important. Realize there is lots of data available—search deep and wide! And talk to other students and faculty.
- **Concepts** (*Optional; due by Sep. 15*) Explore and explain the major theoretical or empirical concepts scholars focus on in your area.
- **Narrowing Your Topics** (*Optional; due by Sep. 22*) Narrow your focus by choosing a few concepts you find interesting. Identify FSU faculty working in these areas. Sketch 3–5 potential research questions.

² See also Part II of Gerring’s *Social Science Methodology* for a supplement to Gerring (2012).